

National Institute of Business Management

Colombo – 07

School of Computing and Engineering

DIPLOMA IN SOFTWARE ENGINEERING

Final Project Proposal Documentation

24.3F

Submission Date – 07/07/2025

Project Title : Website Development for Paradyze (PVT) Limited – An Online Clothing Store

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Name of the Program : Diploma in Software Engineering

Institution : School of Computing and Engineering, National Institute of Business Management

Date : 07/07/2025

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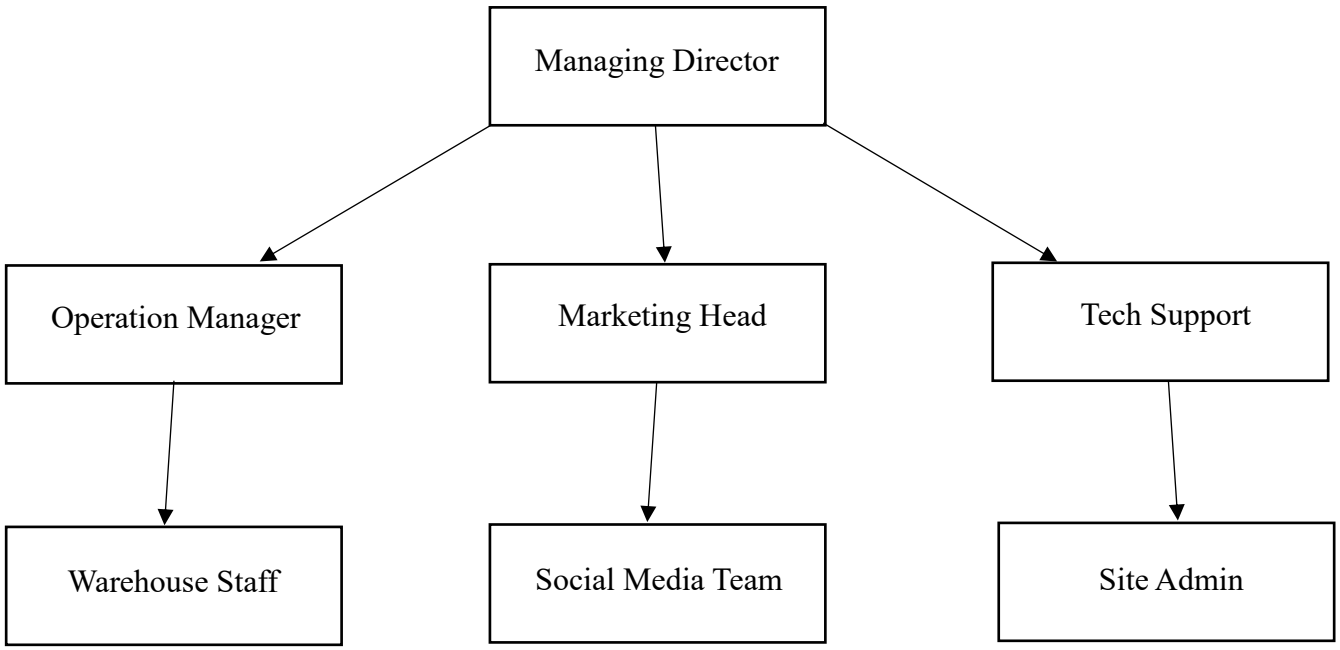
Chapter 1 : Introduction

1.1 Introduction of the Organization

Paradyze (PVT) Limited is a startup in Sri Lanka aiming to redefine the clothing retail experience by operating as a fully online fashion store. With a vision to offer trendy, affordable, and quality fashion items, Paradyze primarily targets young adults and teenagers.

- **Vision :** To become Sri Lanka's most trusted and stylish online fashion destination.
- **Mission :** To provide affordable and fashionable clothing through a fast, reliable, and user friendly e-commerce website

1.2 Organizational Structure



1.3 Current Operations in Organization

Paradyze currently runs its business through social media platforms, mainly Instagram and WhatsApp. Products are showcased through posts and stories, and customers place orders by sending direct messages.

All orders are managed manually using spreadsheets, and stock levels are tracked by staff based on incoming messages and product availability. There is no real-time stock monitoring, leading to frequent inventory errors and overselling.

Payments are via bank transfers or mobile apps and order fulfillment is handled using third party delivery services, coordinated manually. Customer records are not stored systematically and the business lacks automation for order tracking, report generation or performance monitoring.

As order volume increases, this manual approach causes inefficiencies and limits the potential for growth.

1.4 Users and Responsibilities in Organization

Managing Director

- Oversees overall business strategy and growth.
- Approves major decisions related to sales, marketing, and operations.
- Monitors performance through system reports.
- Delegates tasks and evaluates departmental performance.

Operation Manager

- Manages day-to-day business operations.
- Oversees order fulfillment and logistics.
- Coordinates between departments for smooth workflow.
- Reviews operational reports and update management.

Marketing Manager

- Manages product promotions and online advertising.
- Plans and executes seasonal campaigns and discounts.
- Uploads product banners, offers, and announcements on the website.
- Tracks performance of campaign via analytics.

Store Manager

- Maintains and updates product inventory in the system.
- Supervises warehouse staff and order packaging.
- Monitors low-stock items and generates restocking alerts.
- Ensures timely dispatch of confirmed orders.

Warehouse Staff

- Picks and packs items based on confirmed orders.
- Verifies the quality before dispatching.
- Assists in stock audits and physical inventory checks.

Social Media Team

- Posts new product collections and campaigns online.
- Manages daily engagement on Instagram and WhatsApp.
- Handles basic customer questions via DMs before redirecting to support.

Customers

- Responds to customer inquiries via chat/email.
- Assists with order tracking, returns, and refunds.
- Manages complaints and escalates issues to managers.
- Provides feedback to improve service quality.

1.5 Problem Definition

Paradyze currently relies on manual processes and social media platforms to run its online clothing business. While this setup has supported the business in its early stages, it now presents several challenges as customer demand and product variety grow. Orders are taken through direct messages and tracked using spreadsheets, which often leads to human errors, miscommunication, and delayed order processing.

Inventory management is another major issue with no real-time system to monitor stock levels. This results in overselling, out-of-stock issues and customer dissatisfaction. Payments and delivery arrangements are handled separately and manually, causing inefficiencies and delays. Additionally, there is no central database to track customer information, order history, or sales analytics limiting the company's ability to make data-driven decisions.

As Paradyze scales up, the lack of an integrated management system affects its operational efficiency, customer experience, and overall growth potential. To address these challenges, a centralized Online Clothing Store Management System is essential for streamlining operations, automating key processes and supporting future expansion.

1.6 Project Objectives

- **Create a Polished, mobile-friendly e-commerce website.**

  To give customers a dependable and convenient shopping experience, create a simple, aesthetically pleasing website that functions flawlessly on PCs, tablets and smartphones.

- **Facilitate secure checkout, product search and customer registration.**

  Using a safe and easy checkout process, let customers register, browse or search products, add items to their cart and finish purchases.

- **Offer an administrative dashboard for stock and order management.**

  Give administrators a centralized dashboard so they can effectively view customer data, add or remove products, track orders in real time, and update inventory levels.

- **A neat, branded user interface will enhance the client experience.**

  Create a user interface that embodies the Paradyze brand while guaranteeing sample navigation, eye-catching product displays and seamless interaction for customers.

- **Automate sales reports, order tracking and inventory updates.**

  Integrate features that automatically generate sales reports, update stock levels after each purchase, and allow customers to track their order status without manual input.

1.7 Proposed Solution

Our development team suggests creating and implementing a centralized online clothing management system to help Paradyze its online clothing business. With back-end management features integrated, this system will function as a fully functional e-commerce platform.

Every step of the business process will be streamlined by the solution, including inventory control, order management, report generation, product listing and customer interaction. Modern, user-friendly technologies will be used in the system's development to guarantee scalability, security and dependability. Through a dedicated back-end interface, administrators and employees will be able to effectively manage business operations while customers will be able to interact with the store via a responsive website.

Functional Requirements:

- **Customer Management Module**
  - User registration and profile management
  - Login/Logout with password reset
  - Order history and tracking
  - Wishlist and review features
- **Product and Inventory Module**
  - Product listing with categories, size and filter options
  - Real-time stock update
  - Low-stock alerts for admin
  - Automated inventory deduction on sale
  - Add product variants
  - Product zoom and image gallery
  - Product size guide popup
  - Admin alerts when stock is low
- **Order Management Module**
  - Add to cart and secure checkout
  - Order confirmation, status update and invoice generation
  - Return/refund request handling
  - Integration with payment gateways
  - Order cancellation request (before dispatch)
- **Admin Dashboard**
  - View total orders, active customers, top-selling products
  - Add/edit/delete products and categories
  - Manage inventory levels and restocking
  - Approve returns and view sales statistics
  - View customer order history

- **Report Generation**
  - Daily/weekly/monthly sales reports
  - Inventory summary reports
  - Customer activity logs
  - Export reports as PDF or Excel
- **Customer Communication Module**
  - Contact form (inquires go to admin email)
  - Email/SMS notifications for orders, status updates
- **Shipping and Delivery Management**
  - Estimated delivery time display
  - Order delivery tracking integration
  - Printable order invoices and delivery slips
- **Supplier Management Module**
  - Add/edit supplier profiles with contact and product info
  - Generate purchase orders for restocking
  - Track supplier deliveries and restock status
  - View supplier order history and performance
- **Employee Management Module**
  - Role based access
  - Staff login with dashboard access based on role
  - Activity logs for user actions
  - Internal system alerts to staff

Non-Functional Requirements:

- **Security**
  - Role-based access control
  - Encryption for secure transactions
  - Password hashing and user data protection
- **Usability**
  - Clean UI for both customers and staff
  - Fully responsive across all devices
- **Performance**
  - Fast loading time for product pages
  - Optimized image handling
  - Real-time database syncing
- **Backup and Recovery**
  - Daily automated backups
  - Quick restore options in case of system failure

The goal of this solution is to streamline and digitize Paradyze's entire business process, increasing efficiency, accuracy and customer satisfaction. Paradyze will be able to expand its business, provide a polished shopping experience and maintain its competitiveness in the online fashion retail sector with the implementation for suggested system.

1.8 Chapter Summary

This chapter gives a thorough rundown of Paradyze's current business operations and the main issues the company is facing as a result of its manual and disjointed processes. It pinpointed particular issues with order management, inventory tracking, customer service and general operational effectiveness.

The chapter addressed these problems by outlining the project goals, which included transforming the company with a centralized, easy navigable Online Clothing Store Management System. The functional and non-functional elements that will solve current constraints and enhance business performance were highlighted in the comprehensive suggested solution.

It is anticipated that the suggested system will improve customer satisfaction, decrease manual labour and streamline operations, eventually allowing Paradyze to grow and successfully compete in the online retail market.

Chapter 2 : Methodology

2.1 Introduction

The approach that will be used in the creation of Paradyze's online clothing store management system is described in this chapter. It outlines the steps the project team will take to collect requirements, schedule the development process, select appropriate tools and technologies and test the system to guarantee dependability and functionality.

While providing flexibility for enhancements throughout the development cycle, the selected methodology is intended to guarantee that the finished product fulfils the expectations of the company and its clients. The team wants to provide a safe, scalable and intuitive solution that tackles all of the business issues that have been identified by using an organized and iterative process.

2.2 Data Collection Methods

Several data collection techniques were employed to obtain precise and pertinent information from the stakeholders in order to guarantee that the suggested Online Clothing Store Management System satisfies Paradyze's needs. By using these techniques, the development team was able to comprehend current processes, pinpoint major issues and ascertain the system's functional and non-functional requirements.

- **Interviews**

  One-on-one discussions were held with the business owner, warehouse staff and customer support personnel to understand their roles, challenges, and expectations from the new system. These interviews provided detailed insights into current order handling, inventory issues and customer service limitations.

- **Observations**

  The development team observed the day-to-day operations of Paradyze, including how orders were received via social media, how spreadsheets were used to manually manage inventory, and how delivery coordination was handled. As a result, the group was able to pinpoint inefficiencies and potential areas for automation.

- **Questionaries**

  Regular customers were given a brief set of structured questions to complete in order to get their opinions on the shopping experience. Customer expectations for product search, checkout, payment options and order tracking were ascertained through the responses.

- **Document Analysis**

  To learn more about how the company manages inventory, handles payments, and interacts with clients, existing documents including sales spreadsheets, order logs and social media exchanges were examined.

All of these techniques combined to assist the team in obtaining the data required to create a system that complements Paradyze's ongoing business practices while incorporating enhancements that boost productivity, scalability and client satisfaction.

2.3 Software Development Model

Our development team has decided that the Agile Software Development Model is the best method for creating the Online Clothing Store Management System because of the dynamic nature of e-commerce and the changing needs of a developing company like Paradyze.

Agile provides an adaptable and iterative framework that facilitates frequent delivery of functional features, client collaboration and ongoing improvement. This model works especially well for projects where system requirements may change over time and user input is critical.

Why Agile is suitable for this project?

- **Flexibility and Iterative Development**

  The system can be developed in incremental stages, allowing the team to release usable features early, gathers feedback and make improvements continuously.

- **Client Involvement**

  Regular feedback from Paradyze's staff and potential users ensures that the final product closely aligns with real business needs and user expectations.

- **Risk Reduction**

  Issues can be identified and resolved at early stages in the development cycle, reducing the chances of failure at the final delivery stage.

- **Improved Collaboration**

  Agile promotes constant communication between team members and stakeholders, ensuring transparency, accountability and adaptability throughout the project.

- **Continuous Testing and Delivery**

  Each iteration includes testing and review, which helps maintain high software quality and ensures that functionality is reliable and ready to deploy.

By following the Agile Model, the development team will be able to deliver a high-quality, user-focused system in a structured yet flexible manner, ensuring it meets the operational needs of Paradyze while maintaining room for scalability and enhancements in future

iterations.


AGILE METHODOLOGY
Deploy
Test
Review
Agile Methodology
Develop
Plan
Design
Launch

2.4 Software Development Tools

The project team will make use of a number of cutting-edge and dependable tools and technologies in order to successfully design, develop and implement the Online Clothing Store Management System for Paradyze. These were chosen due to their scalability, community support, ease of use, compatibility with the project scope and suitability for e-commerce applications.

- **WordPress :** Used to build and manage the website.



- **WooCommerce :** Enables e-commerce functionality.



- **Elementor :** A user friendly drag and drop page builder.



- **MySQL :** Store and manage all dynamic data.



- **Figma :** Create UI/UX wireframes and prototypes.



- **Wordfence Security Plugin :** Ensures protection against common threats.



- **UpdraftPlus :** Automates daily backups.



- **Rank Math :** To improve visibility and enhance user reach.



- **Smush :** Used for image optimization.



- **GitHub :** Maintains version history of the project.

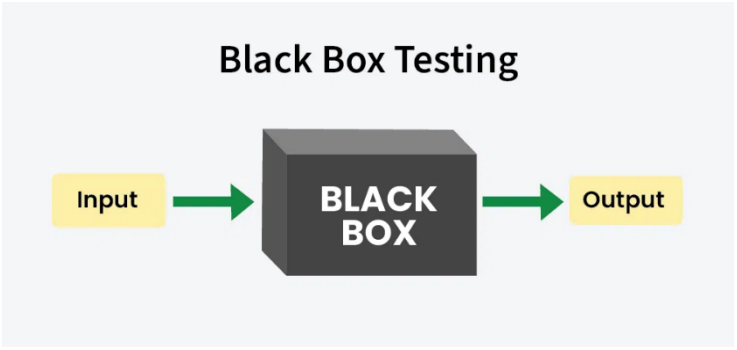


2.5 Testing Strategies

To make sure the Online Clothing Store Management System is dependable, safe and operates as intended, testing is an integral part of the development process. To thoroughly assess the system and guarantee its deployment readiness, the development team will combine Black Box and White Box testing methodologies.

- Black Box Testing

  This focuses on evaluating the system's functionality from the user's perspective, without knowledge of the internal workings. The goal is to test whether the system behaves as expected under real-world conditions.



Functional Testing : This will involve testing each function of the system.

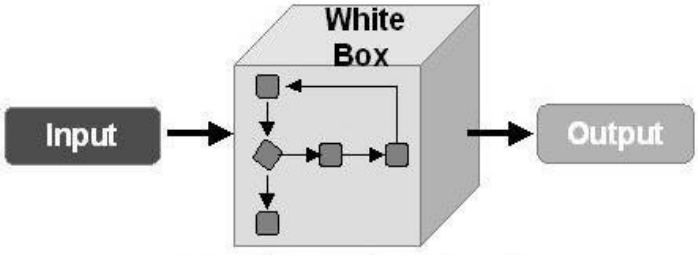
Usability Testing : The system's UI will be tested for ease for use.

Performance Testing : This will be tested under various load conditions to check how it handles multiple users, high traffic and heavy data processing, ensuring optimal performance without delays.

Security Testing : This will focus on ensuring system's security measures.

- White Box Testing

  This involves evaluating the internal workings of the system, including the source code, database interactions and logic flows.



Unit Testing : Each module will be individually tested to ensure that all individual components work as expected.

Integration Testing : This will test how well the different modules of the system work together.

Database Testing : Testing will ensure that the MySQL database interacts correctly with the front-end system.

Regression Testing : After each update or change to the system, this will be performed to ensure that new features do not break existing functionality.

By using these testing strategies the team will ensure that the Online Clothing Store Management System is vetted for functionality, security and performance for Paradyze's growth.

2.6 Implementation Plan

The implementation plan offers a thorough road map for the effective creation and rollout of Paradyze's online clothing store management system. This strategy describes the project's methodical development from the first requirement collection to the last delivery, guaranteeing that every phase is finished on schedule and significantly advances the system's overall goals.

The Agile methodology, which encourages adaptability, ongoing feedback and incremental delivery will be used by the development team. Each weekly phase of the project will concentrate on a different aspect of the system, such as feature development, back-end configuration, user interface design, testing and deployment.

This structured approach allows the team to :

- Identify and resolve issues early
- Adapt to evolving requirements
- Maintain steady progress
- Ensure high system quality and user satisfaction

To verify system performance, security and functionality, frequent testing and stakeholder reviews will be carried out during the implementation. In order to guarantee the platform's long-term dependability, the plan also incorporates documentation, optimization and backup plans.

Following this plan will help the development team deliver a fully functional, secure and user-friendly management system that satisfies Paradyze's operational requirement and facilities future business growth.

2.7 Chapter Summary

The approach used to create Paradyze's online clothing store management system was described in this chapter. It started by outlining how crucial it is to employ an organized strategy in order to guarantee the project's successful completion.

To demonstrate how the system requirements were precisely obtained, a variety of data collection techniques were covered, including observations, questionnaires, interviews and document analysis. Because of its adaptability, iterative nature and responsiveness to feedback, the Agile Software Development Model was determined to be the best strategy for this project.

Additionally, the chapter introduced the technologies and tools that were chosen for the purpose of creating a scalable and user-friendly system. These included WordPress, WooCommerce, Elementor, MySQL, Figma and security plugins. In order to guarantee the system's dependability, security and usability, testing strategies encompassing both black box and white box techniques were examined.

A thorough implementation plan that broke down the tasks week by week and explain how each stage adds to the overall development process was finally presented. The team will follow the tactics covered in this chapter to deliver a comprehensive, useful and effective management system that satisfies Paradyze's operational requirements.

PARADYZE (PRIVATE) LIMITED

No. 26, Mallika Mawatha, Off Templars Road, Mount Lavinia | 0112686837

Paradyze (Private) Limited
No.26, Mallika Mawatha,
Off Templars Road,
Mount Lavinia.

01 July 2025

G.C. Wickramasinghe
Director – School of Computing & Engineering
National Institute of Business Management (NIBM)
No. 120/5, Vidya (Wijerama) Mawatha,
Colombo 07,
Sri Lanka.

Dear Madam,

PERMISSION FOR SOFTWARE DEVELOPMENT PROJECT

We acknowledge receipt of your letter dated 26 June 2025 requesting permission for a group of students from the Diploma in Software Engineering Programme at NIBM to conduct their Software Development Project at our organization.

We are pleased to inform you that Paradyze (Private) Limited is happy to grant permission for the mentioned students to carry out their project at our premises.

We understand the purpose of this project and will support the students in gathering the necessary data and information, and provide them with reasonable access to relevant departments within the organization, under supervision.

Please let us know of any specific requirements or reporting protocols you wish the students to follow during their project period.

We look forward to collaborating with NIBM and supporting this valuable academic initiative.

Yours faithfully,

.....

W.H.M.A.I. Bandara
(Director)
Paradyze (Private) Limited